

$$f_X(x) = \frac{x^{n/2-1} e^{-x/2}}{2^{n/2} \Gamma(n/2)}$$

$$\phi_X(t) = E(e^{tx}) = \int_{-\infty}^{\infty} \frac{e^{tx} x^{n/2-1} e^{-x/2}}{2^{n/2} \Gamma(n/2)} dx$$

$$= \left(2^{n/2} \Gamma(n/2) \right) \int_{-\infty}^{\infty} x^{n/2-1} e^{-(\frac{1}{2}-t)x} dx \quad \begin{matrix} \rightarrow y \\ dy = (\frac{1}{2}-t) dx \\ x = \frac{y}{\frac{1}{2}-t} \end{matrix}$$

$$= C \int_{-\infty}^{\infty} \left(\frac{y}{\frac{1}{2}-t} \right)^{n/2-1} e^{-y} \frac{dy}{\frac{1}{2}-t}$$

$$= C \int_{-\infty}^{\infty} y^{n/2-1} e^{-y} \underbrace{\left(\frac{1}{2}-t \right)^{-n/2}}_{\text{constant}} dy$$

$$= \left(2^{n/2} \Gamma(n/2) \right)^{-1} \left(\frac{1}{2}-t \right)^{-n/2} \left(\int_{-\infty}^{\infty} y^{n/2-1} e^{-y} dy \right)$$

$$= \cancel{2^{n/2}}^{-1} (1-2t)^{-n/2} \cancel{2^{n/2}} \underbrace{\Gamma(n/2)}_{\text{cancels out}}$$

The integral can be written with limits 0 to ∞ as $X \geq 0$